
Coarsened Causal Bayesian Optimization

draft skeleton

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Abstract

1 Introduction

Contributions.

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2 Background

SCMs.

Causal Bayesian Optimization.

$$(S^*, \mathbf{x}^*) = \arg \min_{\substack{S \in \mathfrak{S} \\ \mathbf{x}_S \in \mathcal{X}_S}} \mathbb{E}[Y \mid (X_S = \mathbf{x}_S)]. \quad (1)$$

Coarsening and C-DAGs.

Definition 1 (Coarsening [Madaleno et al., 2026]). *Given a DAG $\mathcal{G}$ on $\mathbf{V}$, a **coarsening** is a partition $\pi = \{C_1, \dots, C_{d'}\}$ such that the induced quotient $\mathcal{G}^\pi$, with nodes $\{C_1, \dots, C_{d'}\}$ and $C_k \rightarrow C_\ell$ iff $\exists V_i \in C_k, V_j \in C_\ell$ with $V_i \rightarrow V_j$ in $\mathcal{G}$, is a DAG.*

3 Coarsened Causal Bayesian Optimization

3.1 Setup

$$\mathfrak{S}^\pi = \left\{ \bigcup_{C_k \in \mathcal{S}} (C_k \cap \mathbf{X}) \mid \emptyset \neq \mathcal{S} \subseteq \mathbf{X}^\pi, |\mathcal{S}| \leq m \right\}. \quad (2)$$

Remark 1 (Intervention granularity).

3.2 Two-tier causal GP prior

$$\mu_0^S(\mathbf{x}_S) = \begin{cases} \hat{\mathbb{E}}[Y \mid (X_S = \mathbf{x}_S)] & \text{if identifiable from } \mathcal{G}^\pi, \\ \bar{Y}_{\text{obs}} & \text{otherwise.} \end{cases} \quad (3)$$

Table 1: Identification method on $\mathcal{G}^\pi$ for each intervention set, CompleteGraph. $\pi_\top$ = identity; $\{B, D\}, \{E\}$ = manip. cluster.

Partition	Intervention set S	Method on $\mathcal{G}^\pi$	Details
$\pi_\top$ (identity)	$\{D\}$	Backdoor	$\mathbf{Z} = \{C\}$
	$\{E\}$	Backdoor	$\mathbf{Z} = \{A, C\}$
	$\{B\}$	Frontdoor	$\mathbf{M} = \{C\}$
	$\{D, E\}$	Backdoor	$\mathbf{Z} = \{A, B\}$
	$\{B, D\}$	G-computation	Propagate through $\{C\}, \{E\}$
	$\{B, E\}$	G-computation	Propagate through $\{C\}, \{D\}$
	$\{B, D, E\}$	G-computation	Propagate through $\{C\}$
$\{B, D\}, \{E\}$ (manip. cluster)	$\{E\}$	Backdoor	$\mathbf{Z} = \{A, C\}$
	$\{B, D\}$	Uninformative	Bidirected edge $\{B, D\} \leftrightarrow Y$ on $\mathcal{G}^\pi$
	$\{B, D, E\}$	Uninformative	Same; E does not break confounding

Algorithm 1 CCBO: Coarsened Causal Bayesian Optimization

Require: $\mathcal{G}^\pi$ (ADMG with bidirected edges for hidden confounders); π ; target Y ; manipulative set $\mathbf{X}$; budget T ; observational data.

- 1: Compute $\mathbf{X}^\pi$ and exploration set $\mathfrak{S}^\pi$ via Eq. (2).
 - 2: For each $S \in \mathfrak{S}^\pi$, call ID (Shpitser & Pearl 2006) on $\mathcal{G}^\pi$ via `ananke-causal`; compute μ_0^S via Eq. (3); fall back to $\bar{Y}_{\text{obs}}$ if NOTIDENTIFIED.
 - 3: Collect initial interventional data $\mathcal{D}_0$.
 - 4: **for** $t = 1, \dots, T$ **do**
 - 5: Update GP posteriors (μ_t^S, k_t^S) for all $S \in \mathfrak{S}^\pi$.
 - 6: $(S_t, \mathbf{x}_t) \leftarrow \arg \max_{S, \mathbf{x}} \text{EI}_{S,t}(\mathbf{x})$.
 - 7: Observe y_t ; update $\mathcal{D}_t$.
 - 8: **end for**
 - 9: **return** $\mathbf{x}_{t^*}$ where $t^* = \arg \min_t y_t$.
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3.3 Assumptions

Assumption 1 (Valid partition).

Assumption 2 (Cluster-level identification as subset prior).

Assumption 3 (GID-PO completeness boundary).

3.4 Algorithm

4 Theoretical properties

Proposition 1 (Identity recovery). *When $\pi = \pi_\top$, CCBO reduces to CBO with power-set exploration.*

Proposition 2 (Identification invariance under intra-cluster edges). *Let π be a valid coarsening of $\mathcal{G}$ and $e = (V_i \rightarrow V_j)$ with $V_i, V_j \in C_k \in \pi$. Then (i) $\mathcal{G}^\pi = (\mathcal{G} \setminus e)^\pi$, and (ii) for every $S \in \mathfrak{S}^\pi$, the identification formula for $\mathbb{E}[Y \mid (X_S = \mathbf{x}_S)]$ derived from $\mathcal{G}^\pi$ is the same under $\mathcal{G}$ and $\mathcal{G} \setminus e$.*

Proof. □

Remark 2 (Structural robustness).

Proposition 3 (Refinement monotonicity). *Let π' refine π . Then (i) π' is valid; (ii) $\mathfrak{S}^{\pi'} \supseteq \mathfrak{S}^\pi$ in the sense that every entry in $\mathfrak{S}^\pi$ maps to at least one entry in $\mathfrak{S}^{\pi'}$; (iii) every effect identifiable from $\mathcal{G}^\pi$ remains identifiable from $\mathcal{G}^{\pi'}$.*

Proof. □

Algorithm 2 RCCBO: Recursive Coarsened Causal Bayesian Optimization

Require: Manipulative $\mathbf{X}$, $\mathcal{D}_{\text{obs}}$, budget T , phase k , RePaRe levels α, β .

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1:  $\pi \leftarrow \text{None}$ ;  $\mathfrak{S} \leftarrow \{\{x\} : x \in \mathbf{X}\}$ ;  $\mathcal{D}_{\text{int}} \leftarrow \emptyset$ .
2: while budget remaining do
3:   if  $\pi \neq \text{None}$  then
4:     Build  $\mathcal{G}^\pi$ ; compute  $\mathfrak{S}^\pi$  and two-tier priors
5:   else
6:     Uninformative priors
7:   end if
8:   Run  $\min(k, \text{rem.})$  trials of CBO on  $\mathfrak{S}$ ; record full-variable samples into  $\mathcal{D}_{\text{int}}$ .
9:    $\pi' \leftarrow \text{REPARE}(\mathcal{D}_{\text{obs}}, \mathcal{D}_{\text{int}}; \alpha, \beta)$ .
10:  if  $\pi' \neq \pi$  then
11:    Patch GP state;  $\pi \leftarrow \pi'$ .
12:  end if
13: end while
14: return  $\mathbf{x}_{t^*}$ , final  $\pi$ .
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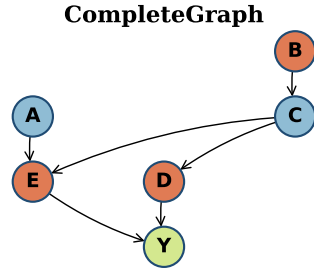


Figure 1: Causal DAG (CompleteGraph). Orange = manipulative; green = target; bidirected $\leftrightarrow$ = latent confounder.

5 Recursive CCBO (teaser)

6 Related work

Table 2: Positioning of CCBO/RCCBO vs. nearest-neighbor methods.

Method	What is learned	Prior assumed	Search space
GACBO [Mukherjee et al., 2024]	Posterior over full DAGs	Uniform DAG prior	$\mathbf{M}$
Durand 2025	$\text{Pa}(Y) \subseteq \mathbf{M}$	Y has no downstream effect	$\mathbf{M}$
Park 2025 [Park et al., 2025]	$\text{PAG} \rightarrow \text{MAG}$ refinement	MEC known	Partial $\mathbf{M}$
CBO [Aglietti et al., 2020]	Nothing (oracle DAG)	Full DAG	POMIS
CCBO (ours)	Nothing (partition given)	Partition of $\mathbf{M}$	POMIS on $\mathcal{G}^\pi$
RCCBO (ours)	Partition π of $\mathbf{M}$	None (start coarsest)	Monotone refines to CBO

Table 3: Final best Y (mean $\pm$ std, 5 seeds $\times$ 40 trials) per method and partition.

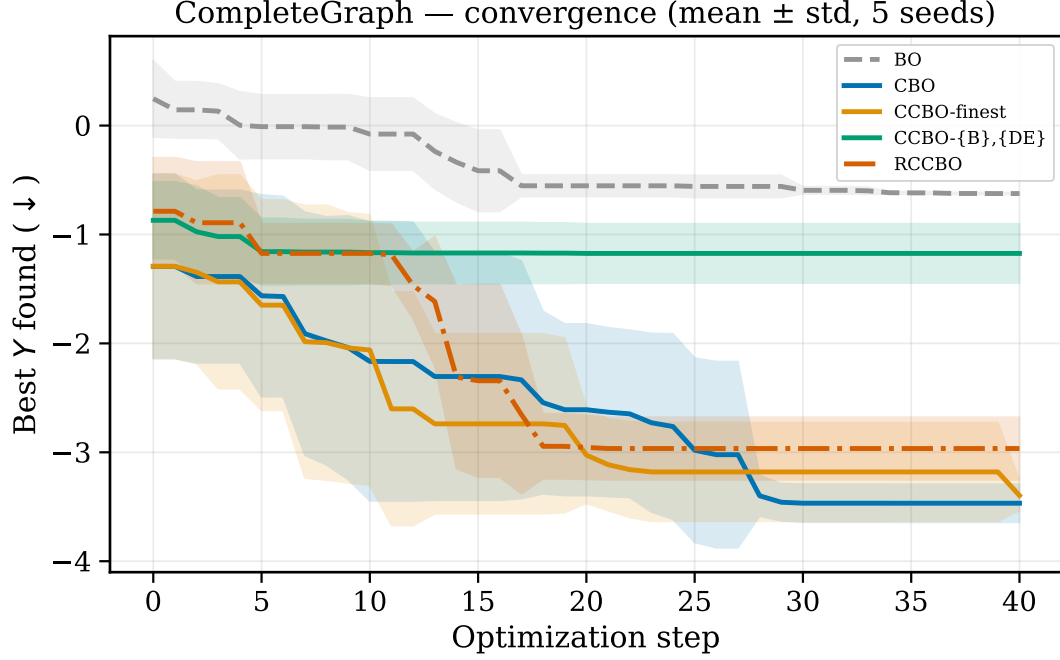


Figure 2: Convergence on CompleteGraph (5 seeds $\times$ 40 trials, mean $\pm$ std). CCBO-finet tracks CBO; CCBO- $\{B\}$, $\{D, E\}$ collapses to uninformative prior.

7 Experiments

7.1 Setup

7.2 Main result

7.3 Robustness (Prop. 2)

7.4 RCCBO teaser

8 Discussion

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9 Limitations

10 Conclusion

References

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Table 4: Prop. 2 validated across two misspec types and two benchmarks.

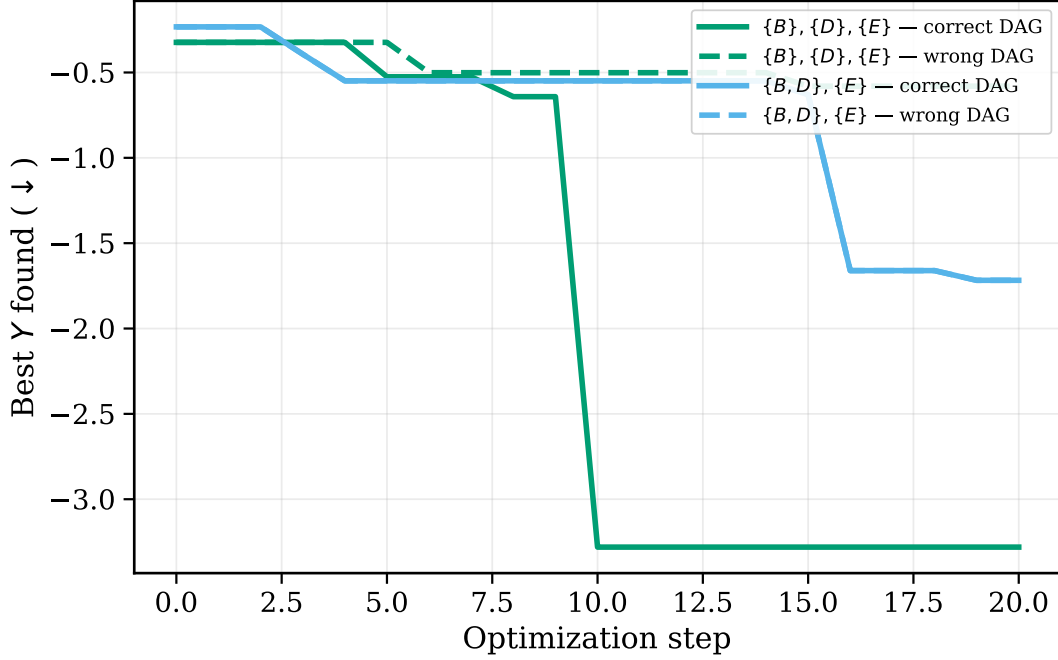


Figure 3: Robustness: correct (solid) vs. misspecified (dashed) DAG. Intra-cluster misspec leaves CCBO unchanged (Prop. 2).

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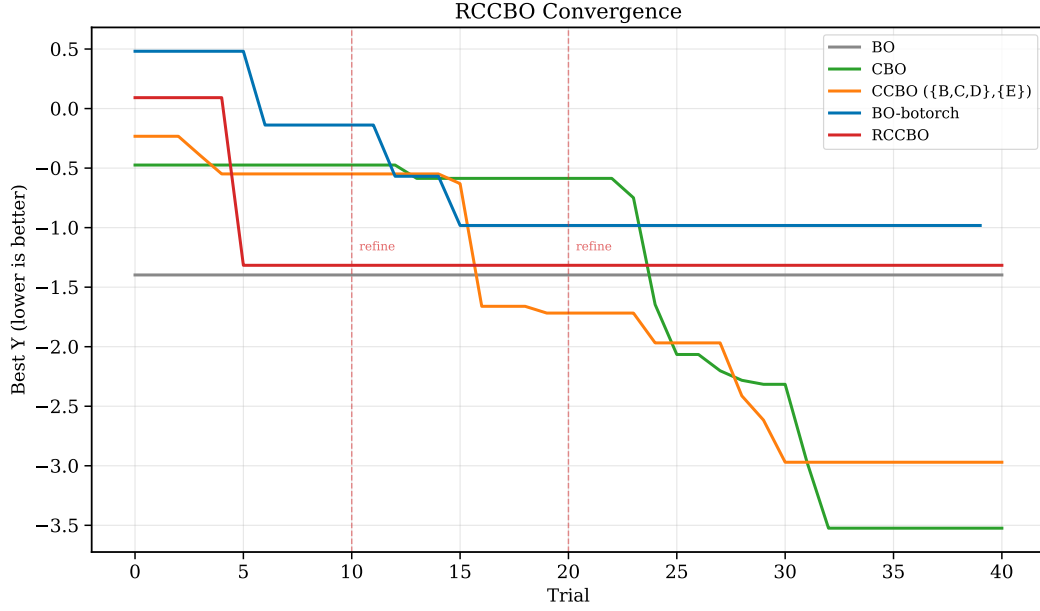


Figure 4: RCCBO: convergence (top) with dashed refinement markers and partition-evolution trace (bottom).

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